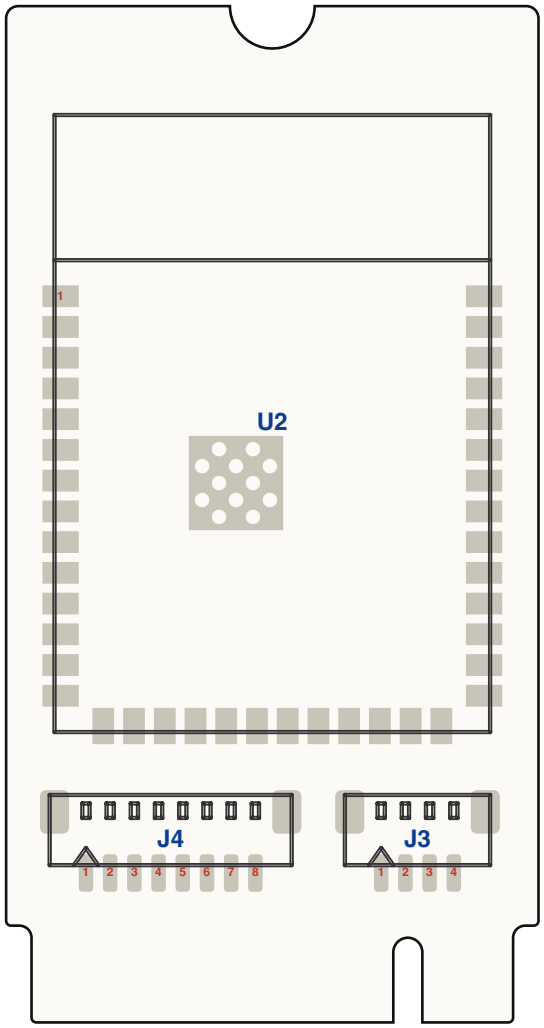
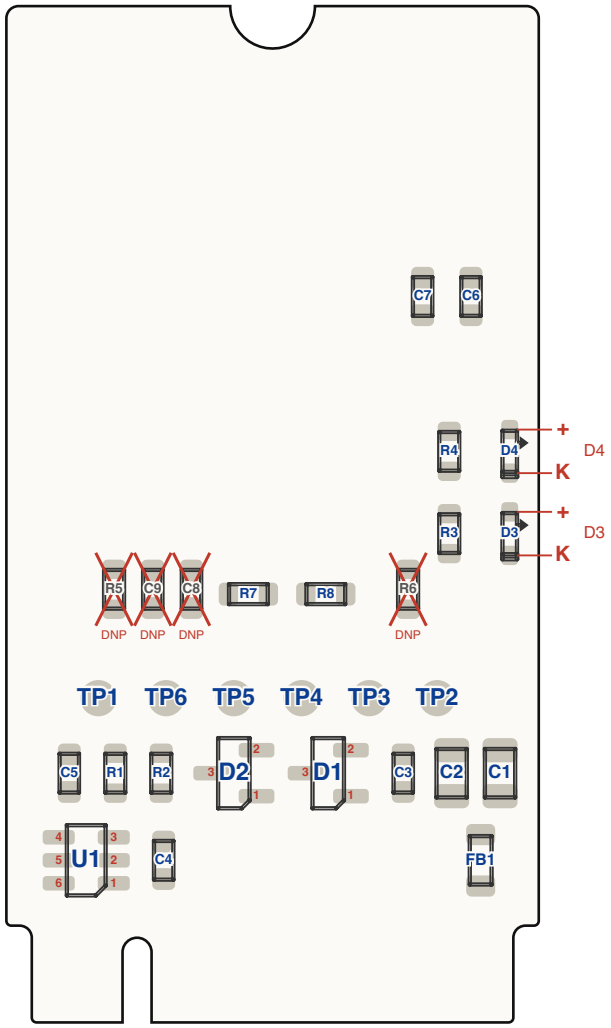
TOP



BOTTOM — seen from below



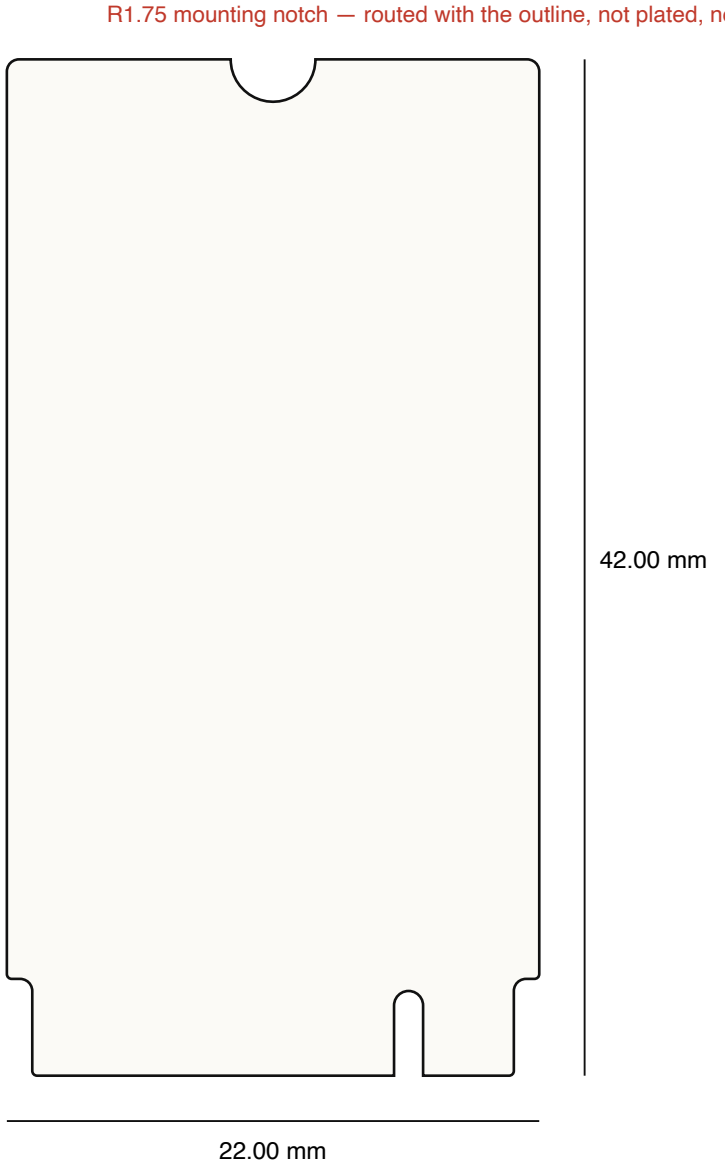
Scale 3.2:1. Grey = copper pads; bold = refdes.
Red numbers = pad numbers (polarized parts, connectors).
LEDs: '+' = anode (pad 2, wired to R3/R4),
'K' = cathode (pad 1) = SILK BAR end.
D1/D2 BAT54A: pin 3 = common anode (card side).
U1 USBLC6: pin 1 = I/O1 (D+ in), silk triangle.
J3 pin 1 = GND (Qwiic); J4 pin 1 = TXD0.
DNP (red X): R5 R6 (I2C pull-ups), C8 C9 (USB SI).
TP1-6: bare pads, not placed (silk: function).
Bottom view mirrored like JLC's bottom preview:
N/S (screw / fingers) is invariant, E/W swap.
Verify every polarized part against the JLC
preview with fab/polarity_checklist.md.

ESP32-M2-COMPANION — assembly

Sheet 1/2 · assembly drawing

Rev v1.0 Rev B · v1.0-34-g16aa571

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Hole table

PTH Ø0.30 mm × 12

PTH via Ø0.30 mm × 45

4 layers SIG/GND/PWR/SIG, 0.80 mm, stackup JLC04081H-3313
ENIG; hard-gold fingers; NO bevel (board < 50 mm):
hand-chamfer the leading edge before first insertion
Mounting notch: routed in the outline - no drill,
no plating: do NOT select castellated holes
Min track/space 0.127 mm (5 mil), min drill 0.30 mm
Vias tented both sides
Assembly: both sides; 22 placed parts; DNP R5 R6 C8 C9
CPL: D3/D4 +180° vs KiCad (JLC_ROT) - VERIFY PREVIEW

ESP32-M2-COMPANION — fabrication

Sheet 2/2 · fab drawing

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